

MLA0103-Artificial Intelligence with Expert Systems

Assesement Tool

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AI-Based Student Support Assistant using Google Dialogflow ES

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Problem Statement

Universities receive a large number of repetitive queries from students regarding course registration, class timetables, examination schedules, attendance details, library timings, hostel facilities, and other campus services. Students often have to contact faculty members or administrative staff to obtain this information, which can result in delays, increased workload for staff, and inconvenience for students. To address this problem, an **AI-based Student Support Assistant** is developed using **Google Dialogflow ES**. The chatbot provides instant, accurate, and automated responses to common student queries at any time, thereby improving communication and reducing administrative effort.

Agent Name: Student Assistant

Type of Agent: Conversational AI chatbox

Google Dialogflow:

Google Dialogflow ES uses Natural Language Understanding (NLU) powered by Google's machine learning models. It identifies user intent and extracts entities from natural language queries to generate



appropriate responses.

Environment:

The chatbot operates in an online environment using Google Dialogflow ES. It can later be integrated into university websites, mobile applications, or messaging platforms.

Inputs

The chatbot accepts text-based queries such as:

- How do I register for courses?
- Show my timetable.
- When are the semester exams?
- What is my attendance percentage?
- What are the library timings?

Outputs

The chatbot provides appropriate text responses such as:

- Course registration procedure
- Timetable information
- Examination schedule
- Attendance details

- Campus service information

Sensors

The chatbot receives user input through:

- Text entered by the student in the chat interface

Objectives:

The main objectives of the Student Support Assistant are:

- To provide instant responses to student queries.
- To automate frequently asked questions related to academic services.
- To reduce the workload of university administrative staff.
- To improve the availability of student support services.
- To provide accurate and consistent information 24×7.

Knowledge Base

The chatbot stores information related to:

- Course Registration
- Timetable
- Examination Schedule

- Attendance
- Library Services
- Hostel Information
- Medical Centre
- Transport Services



student_assistant

SAVE

General

Languages

ML Settings

Export and Import

Environments

Speech

Share



used for students to ask about their queries related to university, hostel services and transport

DEFAULT TIME ZONE

(GMT+6:00) Asia/Almaty

Date and time requests are resolved using this timezone if not provided in the API requests.

AGENT AVATAR URI

Define URI to agent avatar that will be used in Web Demo and Google Chat integrations.

GOOGLE PROJECT

Project ID	student-assistant-rgce
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BETA FEATURES

☐ Enable beta features and APIs

Be the first to get access to the newest features and latest APIs. ([Full V2-beta API reference](#))

LOG SETTINGS

☒ Log interactions to Dialogflow

Collect and store user queries. Logging must be enabled in order to use Training, History and Analytics.

☐ Log interactions to Google Cloud



Dialogflow Essentials

Global

student_assistant

en

Intents

Entities

Knowledge ^[beta]

Fulfillment

Integrations

Training

Validation

History

Analytics

Prebuilt Agents

Small Talk

Docs

Trial Free

Dialogflow CX ^[new]

Support

Upgrade

Intents

CREATE INTENT

Search intents

Attendance

campus services

Course Registration

Default Fallback Intent

Default Welcome Intent

Exam Schedule

Timetable

Custom System

Search entities

@ departments

@ semester

@ service

Try it now



Agent

USER SAYS [COPY CURL](#)
course registration

DEFAULT RESPONSE ▼

Course registration starts from August 1. Login to the student portal. Choose your semester. Select available subjects. Click Submit.

CONTEXTS [RESET CONTEXTS](#)

__system_counters__

INTENT
[Course Registration](#)

Try it now



Agent

USER SAYS [COPY CURL](#)
attendance

DEFAULT RESPONSE ▼

Minimum attendance required is 80%. You can check your attendance through the Student Portal.

INTENT
[Attendance](#)



Try it now

Agent

USER SAYS

COPY CURL

library medical helpdesk timimgs

DEFAULT RESPONSE

Campus Services Library : 8 AM – 8 PM Medical
Centre : 24x7 Hostel Office : 9 AM – 5 PM
Transport Helpdesk : 8 AM – 5 PM

INTENT

campus services

ACTION

Not available

Try it now

Agent

USER SAYS

COPY CURL

exams

DEFAULT RESPONSE

The examination timetable will be published two
weeks before exams. Please check the
Examination Portal regularly.

INTENT

Exam Schedule

ACTION

DIAGNOSTIC INFO

Try it now

Agent

USER SAYS

COPY CURL

library medical helpdesk timings

DEFAULT RESPONSE

Campus Services Library : 8 AM – 8 PM Medical
Centre : 24x7 Hostel Office : 9 AM – 5 PM
Transport Helpdesk : 8 AM – 5 PM

INTENT

campus services

ACTION

Not available

DIAGNOSTIC INFO



Try it now



Agent

USER SAYS

hello

COPY CURL



DEFAULT RESPONSE



Hello! Welcome to the Student Support Assistant. I can help you with: • Course Registration • Timetable • Attendance • Examination Schedule • Campus Services How may I assist you today?

INTENT

Default Welcome Intent

ACTION

input.welcome

Try it now



Agent

USER SAYS

timetable

COPY CURL



DEFAULT RESPONSE



You can view your class timetable from the Student Portal. Department → Semester → Timetable

INTENT

Timetable

ACTION

Not available

PARAMETER

VALUE

date-time

Try it now



Agent

USER SAYS

who is virat kohli

COPY CURL



DEFAULT RESPONSE



Sorry, I couldn't understand your question. Can you rephrase it? You can ask about: • Exams • Attendance • Timetable • Registration • Campus Services

CONTEXTS

RESET CONTEXTS

__system_counters__

INTENT

Default Fallback Intent

ACTION

input.unknown



Edit with WPS Office

Dialogflow Essentials

Global

student_assistant

en

Intents

Entities

Knowledge [beta]

Fulfillment

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Dialogflow CX [new]

Support

departments

[SAVE](#)

☒ Define synonyms ? ☐ Regexp entity ? ☒ Allow automated expansion ☐ Fuzzy matching ?

AIML	AIML, B.Tech AIML
CSE	CSE
IT	IT
ECE	ECE
EEE	EEE
CS-BS	CS-BS
BIO TECHNOLOGY	BIO TECHNOLOGY
AIDS	AIDS
CSE-AI	CSE-AI
Click here to edit entry	

[+ Add a row](#)

dialogflow.cloud.google.com/#/agent/student-assistant-rgce/editEntity/6a4d4975-8d69-4474-9aab-84cd9833aed2

Dialogflow Essentials

Global

student_assistant

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Intents

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Knowledge [beta]

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Dialogflow CX [new]

Support

semester

[SAVE](#)

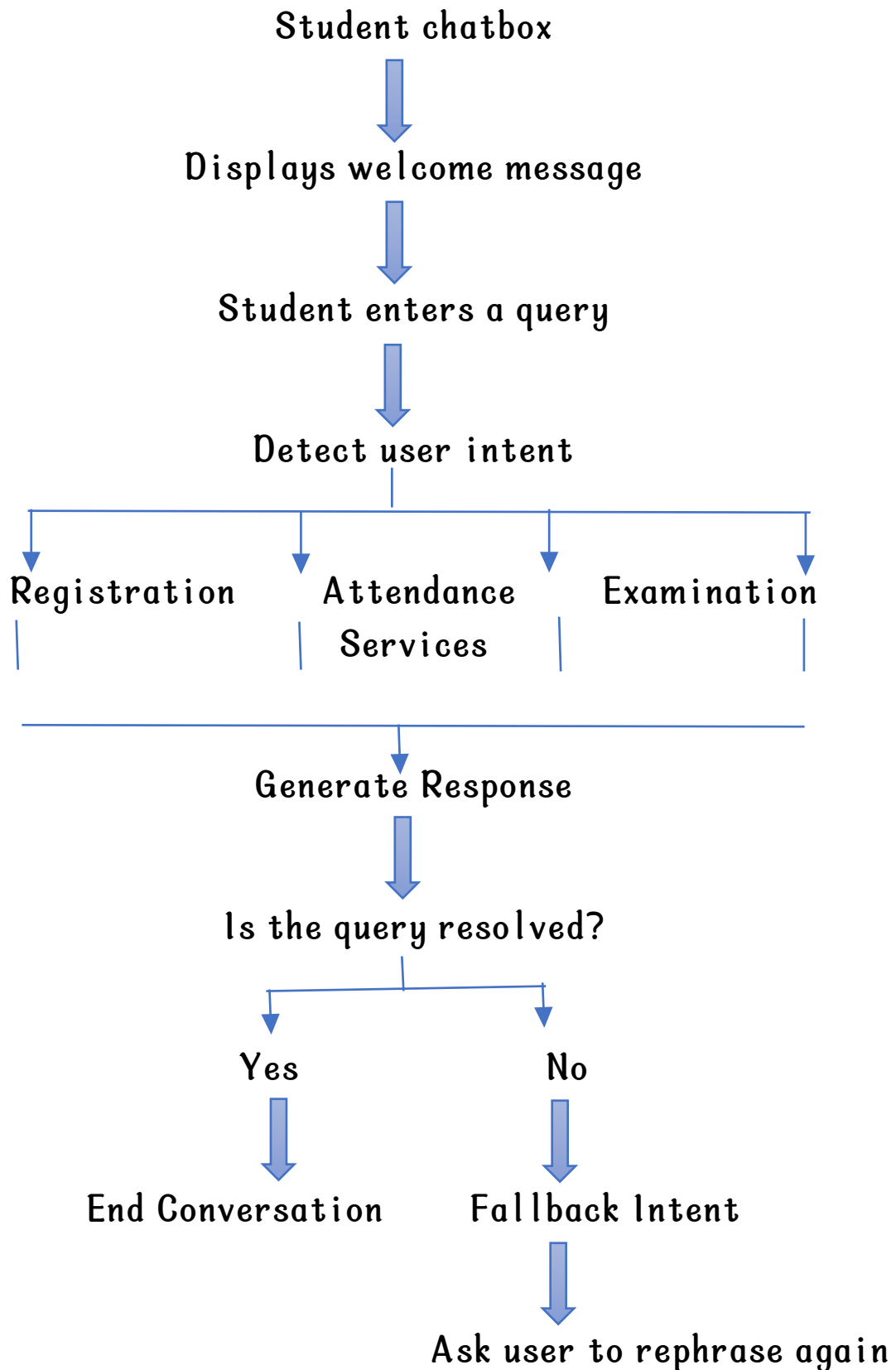
☒ Define synonyms ? ☐ Regexp entity ? ☐ Allow automated expansion ☐ Fuzzy matching ?

1	1
2	2
3	3
4	4
5	5
6	6
7	7
8	8
Click here to edit entry	

[+ Add a row](#)

Edit with WPS Office

Conversation Flow Diagram:



Selected Search Strategy

Heuristic-Based Search:

For this project, **Heuristic-Based Search** is selected because Dialogflow ES uses **Natural Language Understanding (NLU)** and **Machine Learning** to compare the user's query with the available training phrases and identify the most relevant intent.

Instead of checking every intent one by one, the chatbot estimates the best match using confidence scores.

Why Heuristic Search?

Heuristic search is suitable for conversational AI because it:

- Quickly identifies the best matching intent.
- Reduces response time.
- Improves chatbot accuracy.
- Efficiently handles large numbers of intents.
- Provides a better user experience.

Since Dialogflow ES uses Google's machine learning models, heuristic search is naturally integrated into the intent-matching process.

Advantages of the Developed Chatbot

- Provides 24×7 student support.
- Reduces administrative workload.
- Delivers quick and accurate responses.
- Improves communication between students and the university.
- Enhances the overall student experience.

Conclusion:

The **AI-Based Student Support Assistant** was successfully tested using multiple student queries. The chatbot accurately recognized user intents, generated relevant responses, and handled unknown queries using the Default Fallback Intent. The evaluation demonstrated high intent recognition accuracy, fast response time, and reliable performance. With future enhancements such as database integration and multilingual support, the chatbot can become a comprehensive digital assistant for university students.

